

SECTION A [25 marks]

This section consists of **3** questions. Answer **all** questions.

1.

- a) A function $f(a)$ is discontinuous at a for three different situations. State the situations.

[2 marks]

b) Evaluate

i. $\lim_{h \rightarrow 0} \frac{e^{2h} - 1}{e^h - 1}$

[3 marks]

ii. $\lim_{h \rightarrow -\infty} \frac{x - 5}{\sqrt{4x^2 + 3}}$

[4 marks]

2.

- a) $f(x)$ is a quadratic function taking the form $ax^2 + 7x + b$. If $f(-1) = 6$ and $f'(2) = 27$, find the values of a and b .

[4 marks]

- b) Given $f(x) = \sqrt{2x + 3}$. Find $f'(x)$ by using first principle.

[5 marks]

3. Given that $f(x) = x^3 - 3x^2 - 24x + 32$. Find and classify the stationary point(s) by using the first derivative test. Hence, sketch the graph.

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. Given that $z = -3\sqrt{2} - 3i$.

a) Determine the modulus and argument of z .

[3 marks]

b) Express z in polar form.

[2 marks]

2.

a) Solve the equation $\sqrt{4x+9} - \sqrt{2x+1} = 2$

[5 marks]

b) Solve the inequality $\left| \frac{x-4}{x+2} \right| \geq 4$

[6 marks]

3.

a) Given that $f(x) = \frac{1}{\sqrt{3x-7}}, x > \frac{7}{3}$ and $g(x) = 3px^{-2} + q, x > 0$. Find the values of p and q if $f(x)$ and $g(x)$ are inverse each other.

[5 marks]

b) Given two functions, $q(x) = 3x + 5$ and $r^2q(x) = \frac{1}{2} + \frac{1}{3}x$. Find $r(x)$, where $r(x)$ is a linear function.

[5 marks]

4. The function f and g are defined by $f(x) = \ln(8 - 3x)$ and $g(x) = e^{2x-4}$.

a) Determine the inverse function of $f(x)$.

[2 marks]

b) State the domain and range of $f^{-1}(x)$.

[2 marks]

c) Sketch the graphs of f and f^{-1} on the same axis.

[3 marks]

d) Find $g \circ f$.

[3 marks]

5.

a) The function f is defined by

$$f(x) = \begin{cases} \frac{x^2 - 1}{x - 1} & , x < 1 \\ m^2 - 2 & , x = 1 \\ \frac{|1 - x|}{x - 1} + n & , x > 1 \end{cases}$$

where m and n are constants.

i. If $\lim_{x \rightarrow 1} f(x)$ exists, find the value of n .

[5 marks]

ii. If f is continuous at $x = 1$, determine the values of m .

[2 marks]

b) A function f is defined by $f(x) = \frac{x}{x+3}$.

i. Find the vertical asymptotes of f .

[2 marks]

ii. Find the horizontal asymptotes of f .

[2 marks]

iii. Sketch the graph of f and state the domain.

[3 marks]

6.

a) Find the gradient of the curve with equation $4\sin x \cos y + \sec y = 5$ at the

point $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

[8 marks]

b) A curve is defined by the parametric equations:

$$x = \frac{9}{3-t} \quad y = e^{-t}$$

i. Show that $\frac{dy}{dx} = \frac{-(3-t)^2}{9} e^{-t}$

ii. Find an expression for $\frac{dy}{dx}$ in terms of y and x .

iii. Find $\frac{d^2y}{dx^2}$ in terms of $\frac{dy}{dx}$, y and x .

[8 marks]

7. A conical container of base radius 5 cm and height 10 cm is being filled with sand at the rate of $2 \text{ m}^3 / \text{min}$. How fast is the level of the sand rising when it is 6 cm above the apex of the conical container?

[9 marks]

Suggested Answer

Section A

- a) $f(a)$ is not defined, the limit at a does not exist, $\lim_{x \rightarrow a} f(x) \neq f(a)$
 b) i. 2 ii. $-\frac{1}{2}$
- a) $a=5, b=8$
 b) $\frac{1}{\sqrt{2x+3}}$
- $\max(-2, 60) \min(4, -48)$ inflection $(1, 6)$

Section B

- a) $|z|=3\sqrt{3}, \quad \theta=-2.53$
 b) $3\sqrt{3}(\cos(-2.53)+i\sin(-2.53))$
- a) $x=0$ and $x=4$
 b) $\left\{x: -4 \leq x < -2 \text{ or } -2 \leq x < -\frac{4}{5}\right\}$
- a) $p=\frac{1}{9}, \quad q=\frac{7}{3}$
 b) $r(x)=\frac{1}{3}x-\frac{1}{24}$ or $r(x)=-\frac{1}{3}x-\frac{1}{12}$
- a) $f^{-1}(x)=\frac{8-e^x}{3}$
 b) $D_f = R_{f^{-1}} = (-\infty, \frac{8}{3})$
 $R_f = D_{f^{-1}} = (-\infty, \infty)$
 d) $e^{2[\ln(8-3x)]-4}$
- a) i. $n=1$ ii. $m=\pm 2$
 b) i. $x=-3$ ii. $y=1$ iii. $(-\infty, -3) \cup (-3, \infty)$
- a) -1
 b) i. show! ii. $\frac{dy}{dx} = -\frac{9y}{x^2}$ iii. $\frac{d^2y}{dx^2} = \frac{18y}{x^3} - \frac{9}{x^2} \frac{dy}{dx}$
- 0.07074 cm/min